



DD2380 Artificial Intelligence

Probabilistic Reasoning

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Lectures

- Intro and practicalities
- Ethics
- Problem Solving
 - Search
 - Adversarial Search
- **Taming Uncertainty**
 - **Probabilistic Reasoning**
 - Hidden Markov Models (HMMs)
- Invited talk – Multi Agent Systems + Robotics
- Invited talk – Computer Vision
- Making decisions under uncertainty
 - Decision Making
 - Reinforcement Learning (RL)
- Invited talk – Machine Learning (ML)
- Invited talk – Natural Language Processing (NLP) and Conversational AI
- Knowledge, Reasoning, and planning
- Invited talk – Generative AI & Final QA



Reading Instructions

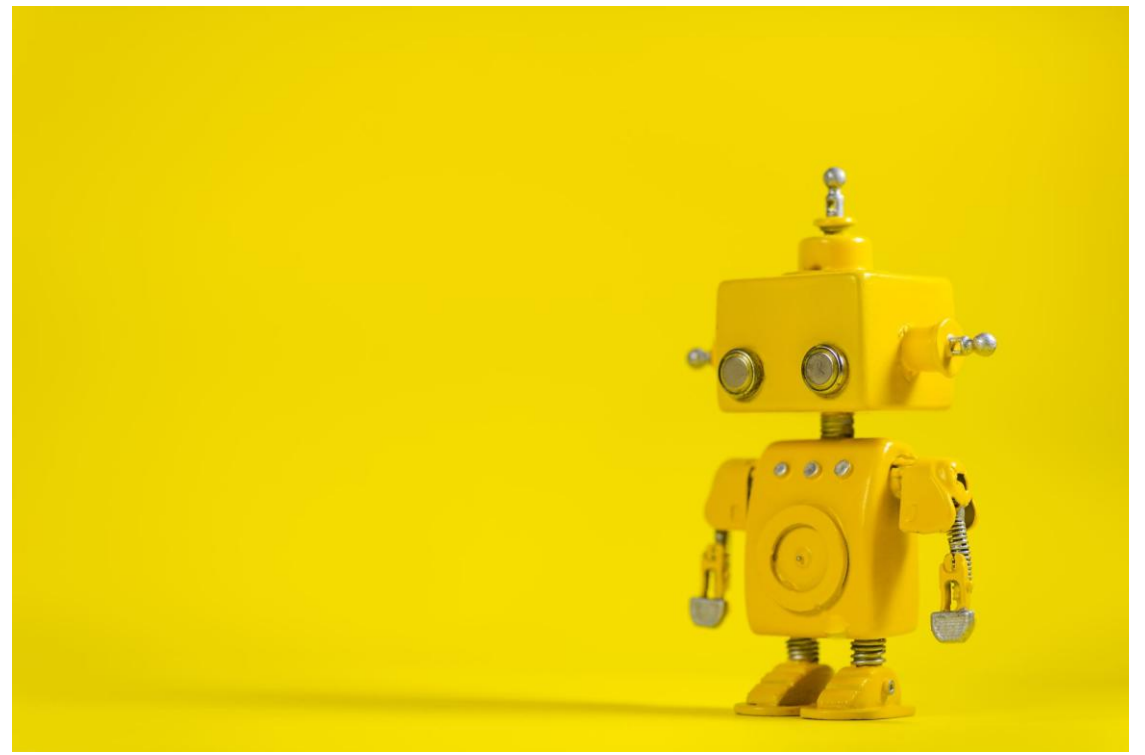
- Chapter 13-14 (3th edition), 12-13 (4th edition), Russell & Norvig AIMA



Outline (Taming uncertainty)

- Probabilities
 - Motivation
 - Notation and Recap
 - Bayes Rule
 - Conditional Independence
- Probabilistic Graphical Models
 - Bayesian Networks
 - Sequential Data
 - Markov Models (next Lecture)
 - Hidden Markov Models (next Lecture)

Motivation: Why We Use Probabilities?



What is Probability/Uncertainty

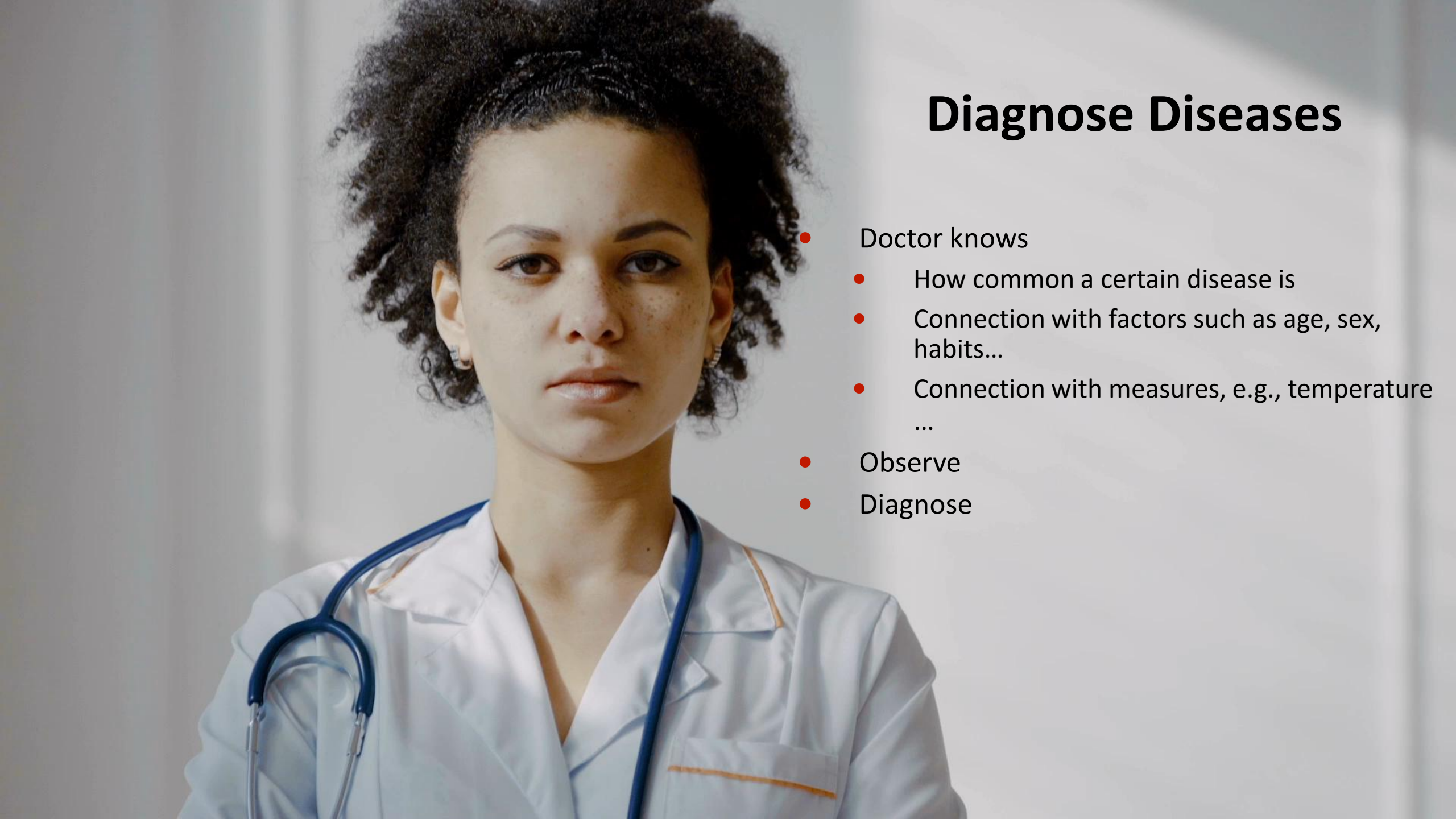
- Probability quantifies the likelihood of an event happening in the face of **uncertainty**.
- Uncertainty plays an important role in
 - sensor interpretation
 - sensor fusion
 - map making
 - path planning
 - self-localization
 - control
 - etc...



Real-World Examples – Autonomous Car

- Cross intersection safely?
 - Observations from car
 - Sensor models
 - Statistics from different roads
 - Weather models
 - ...
 - Observations from others
 - Can I cross with
 - 99% safety?
 - 99.99999% safety?





Diagnose Diseases

- Doctor knows
 - How common a certain disease is
 - Connection with factors such as age, sex, habits...
 - Connection with measures, e.g., temperature...
- Observe
- Diagnose

Pervasive Power of Probabilities



Probability Recap 1/3

- Probability of event X : $p(X)$
- $p(X) \in [0,1]$ (i.e., $0 \leq p(X) \leq 1$)
- $\sum_{\text{all outcomes } X} p(X) = 1$
- $p(\neg X)$ the probability that X is false.
- $p(X) = 1 - p(\neg X)$
- Joint probability of X AND Y : $p(X,Y)$
- Conditional probability of X GIVEN Y : $p(X|Y)$

Probability Recap 2/3

- *Product rule:* $p(X, Y) = p(Y|X)p(X)$
- *Sum rule (marginalization):* $p(X) = \sum_Y p(X, Y)$

Sum Rule (Marginalization)

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

In general: $p(X) = \sum_Y p(X, Y)$

$$p(T) = \sum_W p(T, W)$$

$$p(W) = \sum_T p(T, W)$$

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.4

Marginal distributions

Law of Total Probability (conditioning)

Combining

$$p(X) = \sum_Y p(X, Y) \text{ (sum rule)}$$

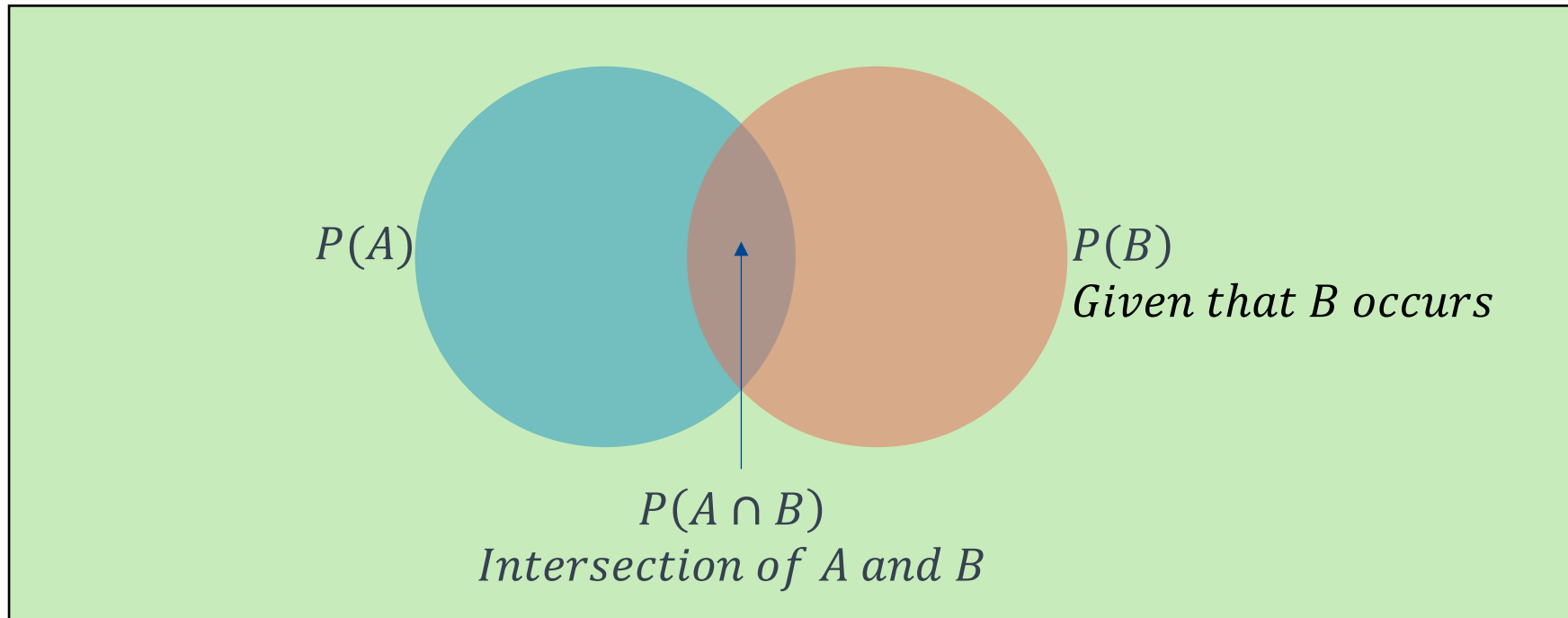
$$p(X, Y) = p(X|Y)p(Y) \text{ (product rule)}$$

Gives

$$p(X) = \sum_Y p(X|Y)p(Y)$$

Conditional Probability

- Given 2 events A and B with $P(B) \neq 0$,
 - the conditional probability of A given B is denoted $P(A|B)$
 - and defined as: $P(A|B) = \frac{P(A \cap B)}{P(B)}$



Conditional Probability (Weather Example)

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

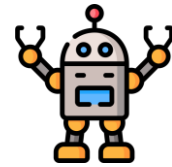
- $P(W = s|T = c) = \frac{P(W=s, T=c)}{P(T=c)} = \frac{0.2}{0.5} = 0.4$

Sum Rule

$$P(W=s, T=c) + P(W=r, T=c) = 0.2 + 0.3 = 0.5$$

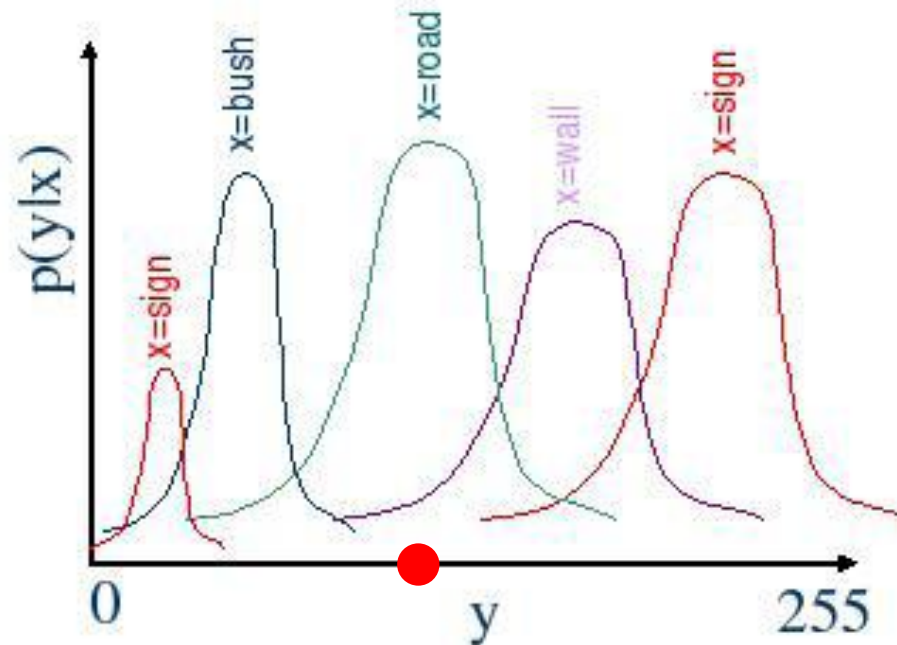
Conditional Dependence

- Applications in Artificial Intelligence
 - **Natural Language Processing:**
 - Understanding pronouns.
 - In the sentence “There's a cat and a dog. John approached it because he heard barking.” The likelihood that "it" refers to "dog" depends on the context provided by "heard barking.”
 - **Robotics**
 - Path planning must consider obstacles.
 - A cluttered environment reduces the likelihood of direct navigation.
 - **Computer Vision**
 - Recognizing street signs.
 - If the system detects a bright backdrop typical of signboards, it also anticipates contrasting dark regions indicative of text or symbols. Recognizing one feature increases the likelihood of the other being present.



Recognizing Street Signs Example

- Knowing what to look gives an idea of what to expect to measure
- We could expect both very bright and very dark pixels to belong to street signs



Probabilistic Inference

- Compute a desired probability from other known probabilities (e.g., conditional from joint)
- **We generally compute conditional probabilities**
 - $P(\text{on time} \mid \text{no reported accidents}) = 0.90$
 - These represent the agent's *beliefs* given the evidence
- **Probabilities change with new evidence:**
 - $P(\text{on time} \mid \text{no reported accidents, 5 a.m.}) = 0.95$
 - $P(\text{on time} \mid \text{no reported accidents, 5 a.m., raining}) = 0.80$
 - Observing new evidence causes *beliefs to be updated*



Bayes Rule

- Allows us to update our beliefs about the probability of an event, given new information
- $$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$
 - $P(A|B)$: Posterior (Probability of A given B is observed)
 - $P(B|A)$: Likelihood (Probability of observing B given A is true)
 - $P(A)$: Prior (Initial probability of A)
 - $P(B)$: Evidence (Overall probability of observing B)
- Why is it useful?
 - Probability of A given some evidence B can be expressed in **factors that are sometimes easier to determine.**
- Foundation of many AI systems



Bayes Rule Derivation (2 different options)

From $P(A|B) = \frac{P(A,B)}{P(B)}$ (conditional probability)

- Use the product rule to replace the joint probability with $P(B|A)P(A)$

Rewrite conditional probability to $P(A,B) = P(A|B)P(B) = P(B|A)P(A)$

- Divide $P(A|B)P(B) = P(B|A)P(A)$ by $P(B)$

To get $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

Bayes Rule using Normalization

$$P(A|B) = \frac{P(B|A)P(A)}{\mathbf{P(B)}}$$

Expressing $P(B)$ by conditioning over A : $P(B) = \sum_A P(B|A)P(A)$

$$P(A|B) = \frac{P(B|A)P(A)}{\sum_A P(B|A)P(A)}$$

$$\text{With } \eta = \frac{1}{\sum_A P(B|A)P(A)},$$

$$\mathbf{P(A|B) = \eta P(B|A)P(A)}$$

AIMA, 3rd edition, p. 493 (section 13.3) for a concrete example

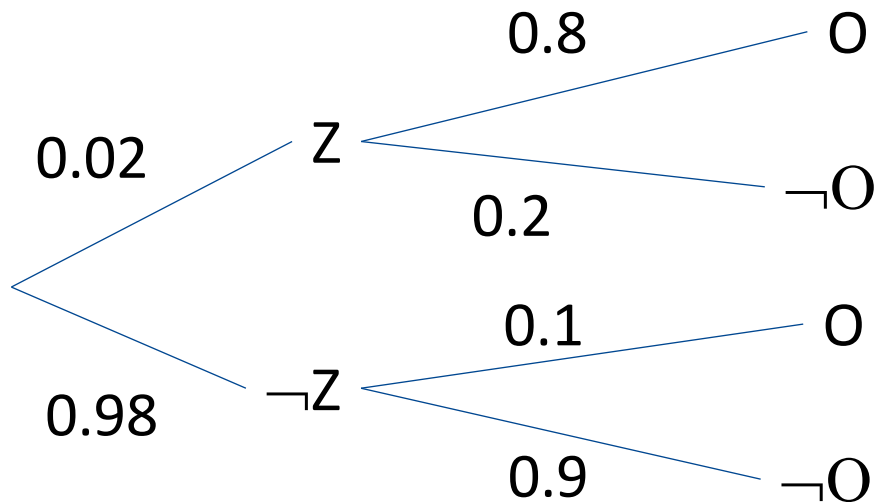
Bayes Rule Example

- Vision system for detecting zebras Z
 - Prior: $p(Z)=0.02$ (zebra in 2% of images)
 - Detector for “stripey areas” gives observations, O
 - Detector gives yes/no
 - Detector performance
 - $p(O|Z) = 0.8$ (true pos)
 - $p(O|\neg Z) = 0.1$ (false pos, e.g. gate)
- Exercise 1: Calculate $p(Z|O)$
 - What does it represent?
 - The result of that probability?



Bayes Rule Example Solution

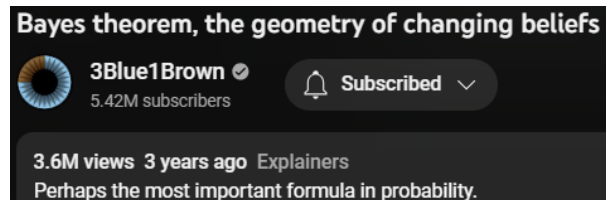
- $p(Z|O)$ represents probability that there is a zebra if our detector says there is one.
- To calculate $P(Z|O) = \frac{P(O|Z)P(Z)}{P(O)}$
 - $p(Z) = 0.02$ (zebra in 2% of images)
 - $p(O|Z) = 0.8$ (true pos)
 - $p(O|\neg Z) = 0.1$ (false pos, e.g., gate)



$$\frac{P(O|Z)P(Z)}{P(O|Z)P(Z)+P(O|\neg Z)P(\neg Z)} = \frac{0.8 \cdot 0.02}{0.8 \cdot 0.02 + 0.1 \cdot 0.98} = 0.1404$$

Bayes Rule Example Discussion

- $p(Z|O) = 0.1404$
- Intuition tells most people that the detector is much better than this, i.e. we would expect to see a much higher $p(Z|O)$ since the detector detects 80% of actual zebras.
 - Among 1,000 images, about 20 contain zebras; the detector flags 16.
 - Among the other 980, it falsely flags 98.
 - Only $16/(16 + 98) \approx 14\%$ of positive detections therefore contain zebras.
- Failing to account for negative evidence properly is a typical failing of human intuitive reasoning



Conditional Independence

- Unconditional (absolute) independence very rare (why?)
- *Conditional independence* is our most basic and robust form of knowledge about uncertain environments.

Example:

P (Traffic)

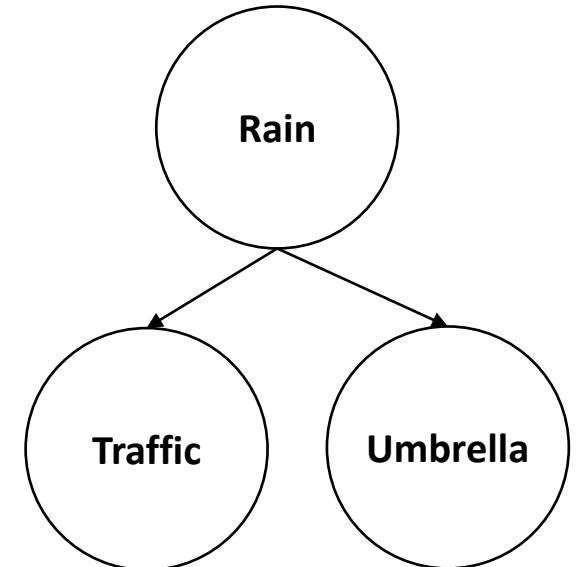
P (Umbrella)

P (Rain)



Rain causes traffic AND people wearing umbrellas.

T is conditionally independent of U given R



Conditional independence Formulas

- If X is conditionally independent of Y given Z

$$P(X|Y, Z) = P(X|Z)$$

- Which also means

$$P(X, Y|Z)$$

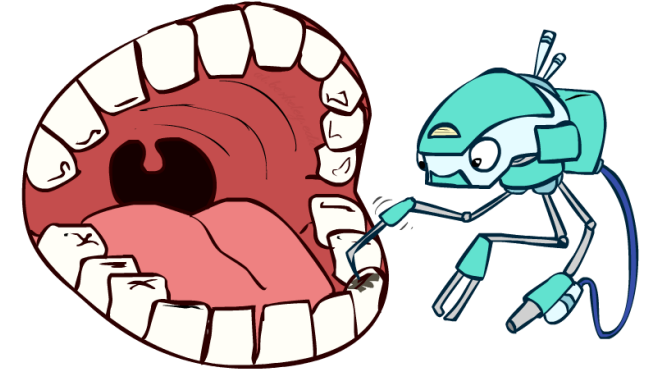
$$= P(X|Y, Z)P(Y|Z) \quad \{\text{product rule: } P(X, Y) = P(X|Y) \times P(Y)\}$$

$$= P(X|Z)P(Y|Z) \quad \{\text{conditional independence}\}$$

- NOTE: Not the same as $P(X, Y) = P(X)P(Y)$

Conditional Independence Example

- $P(\text{Toothache}, \text{Cavity}, \text{Catch})$
- If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache



$$P(\text{Catch} \mid \text{Toothache}, \text{Cavity}) = P(\text{Catch} \mid \text{Cavity})$$

- Catch is conditionally independent of Toothache given Cavity.
- Equivalent statements:

$$P(\text{Toothache} \mid \text{Catch}, \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity})$$

$$P(\text{Toothache}, \text{Catch} \mid \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity}) P(\text{Catch} \mid \text{Cavity})$$

Probability Recap 3/3

Conditional Probability: $P(x|y) = \frac{P(x, y)}{P(y)}$

Product Rule: $p(x, y) = p(y|x)p(x)$

Chain Rule: $P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots = \prod_{i=1}^n P(X_i|X_1, \dots, X_{i-1})$

X,Y independent if and only if $\forall x, y : P(x, y) = P(x)P(y)$

X and Y are conditionally independent given Z if and only if: $X \perp\!\!\!\perp Y | Z$

$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$

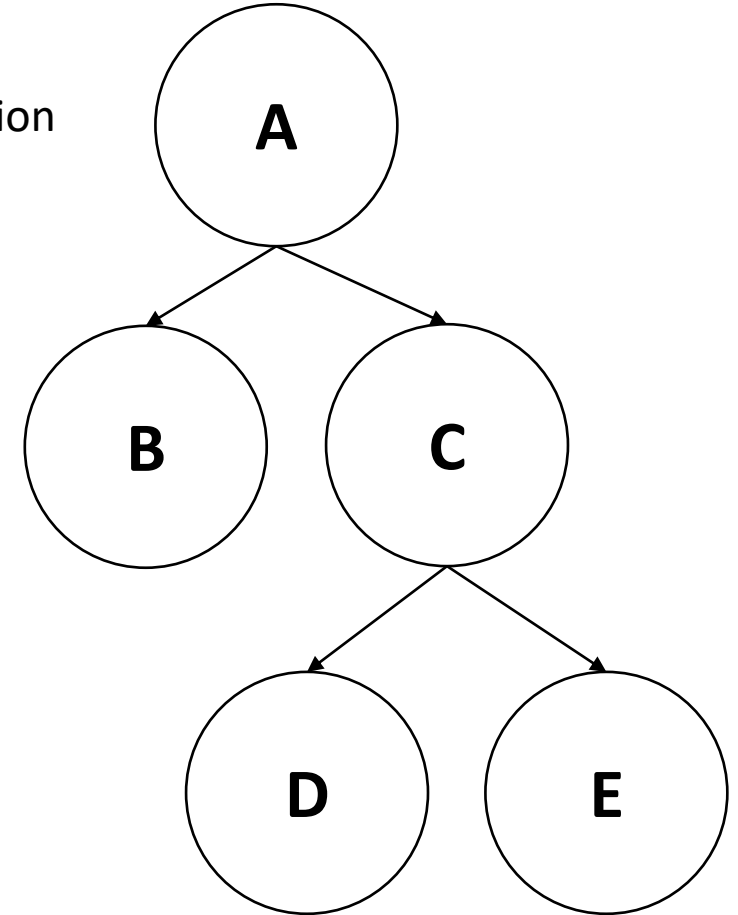


Break



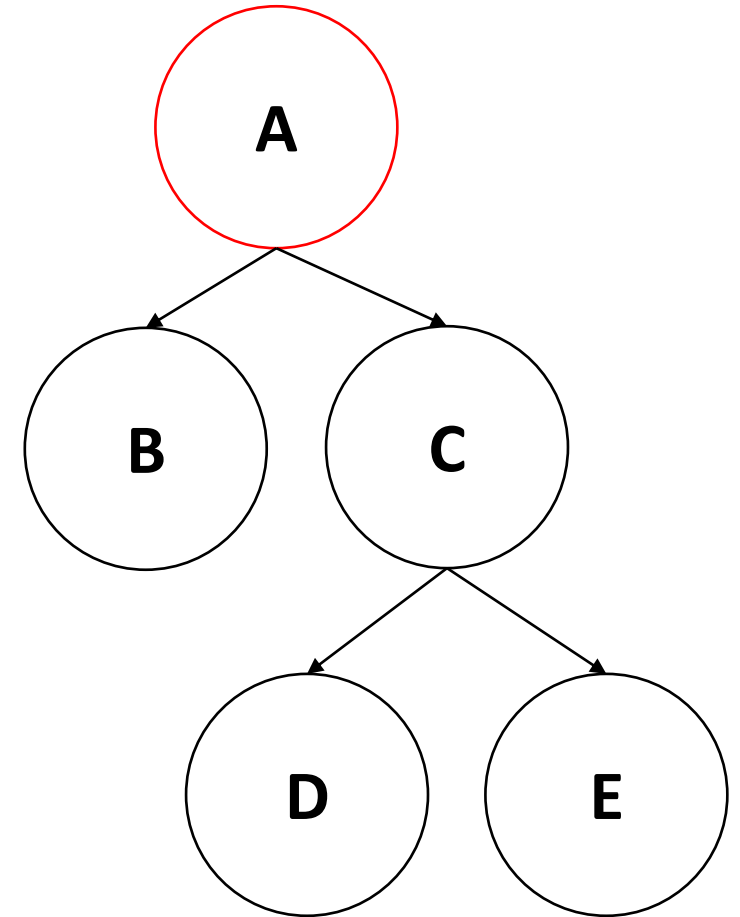
Probabilistic Graphical Models

- Compact representation of the joint distribution over a set of variables
- Graphical representation that helps analyze and structure probability information
- Each variable is encoded as a node
- Conditional independence assumptions coded as arcs
- Here: a Bayesian network
(directed acyclic graph (DAG))



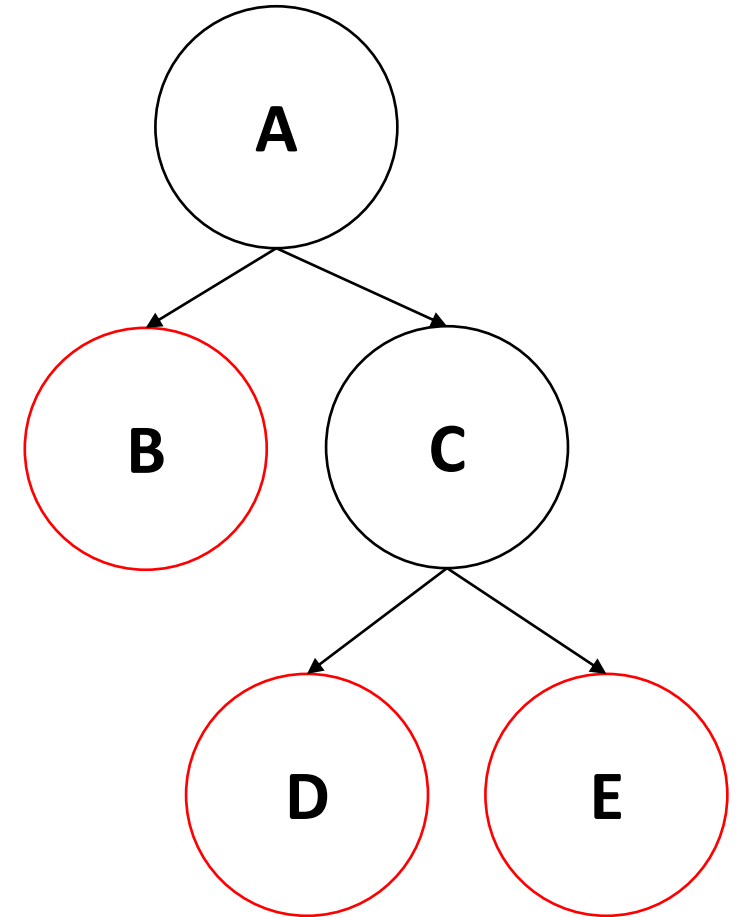
Bayesian Network

- A root node



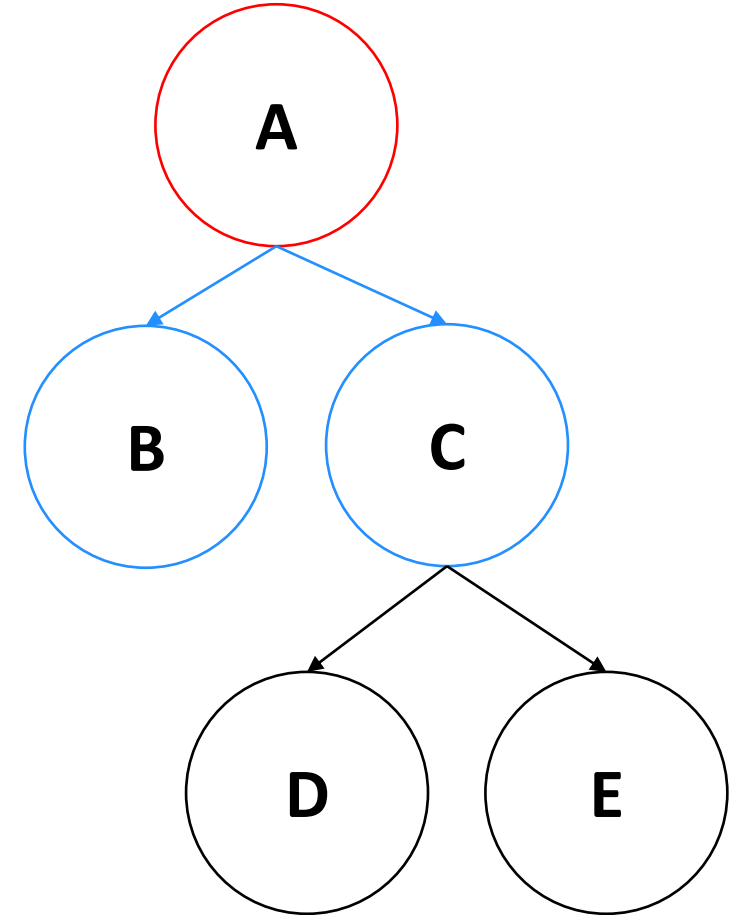
Bayesian Network

- A root node
- B, D, E leaf nodes



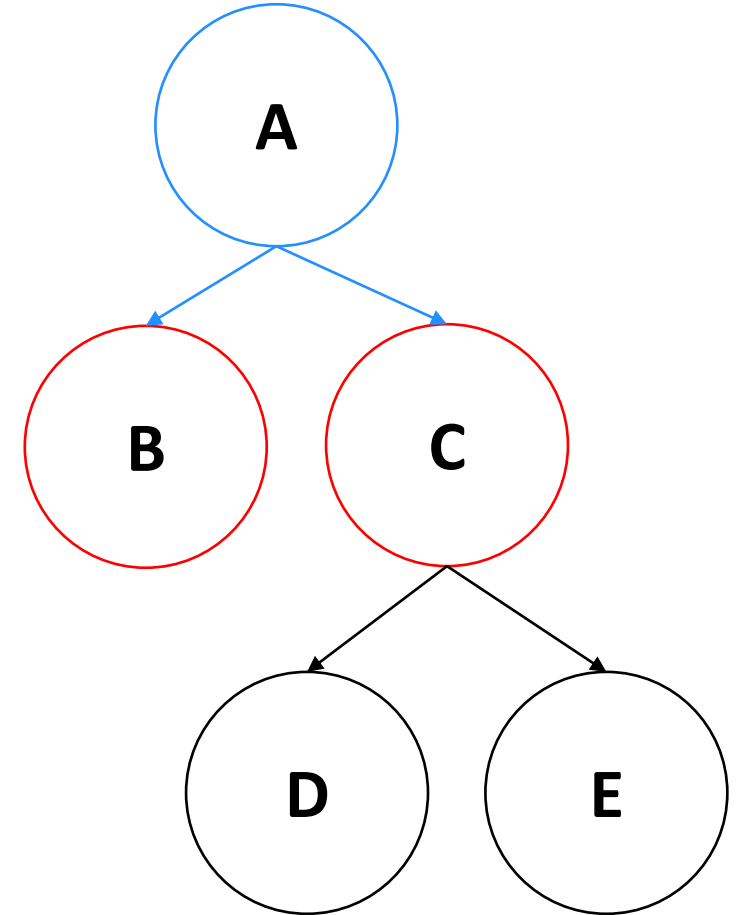
Bayesian Network

- A root node
- B, D, E leaf nodes
- A parent to B and C



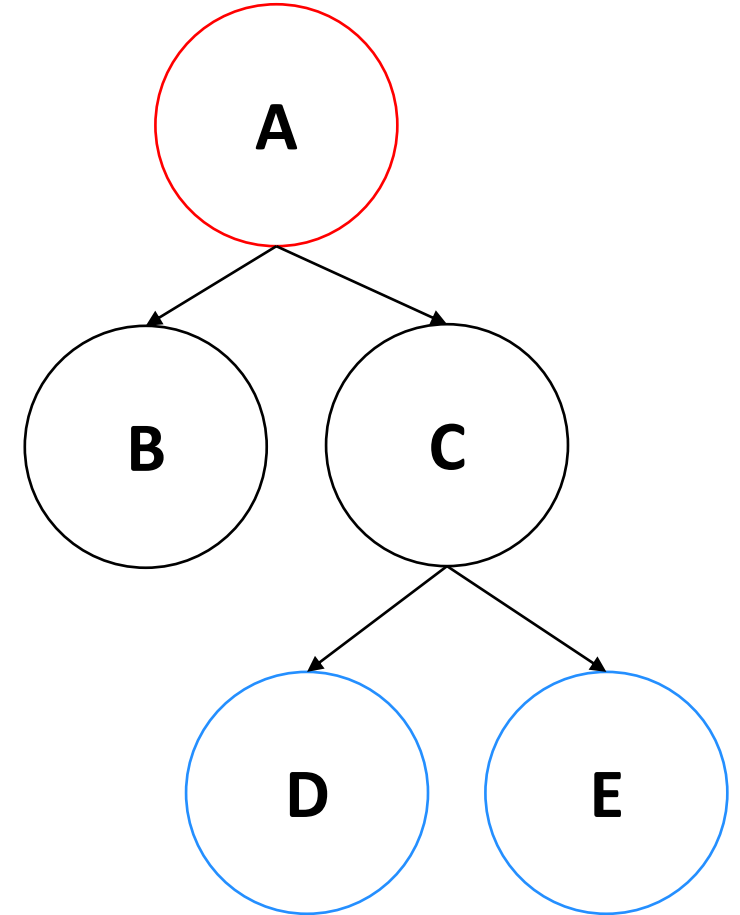
Bayesian Network

- A root node
- B, D, E leaf nodes
- A parent to B and C
- B and C children of A



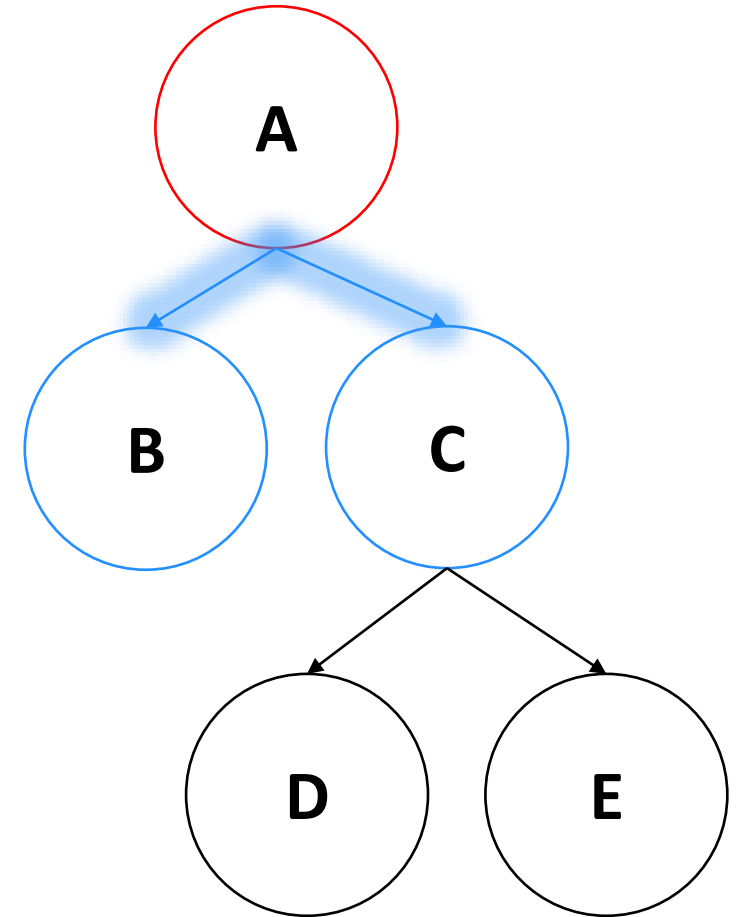
Bayesian Network

- A root node
- B, D, E leaf nodes
- A parent to B and C
- B and C children of A
- A ancestor of D and E



Bayesian Network

- A root node
- B, D, E leaf nodes
- A parent to B and C
- B and C children of A
- A ancestor of D and E
- A “influences” B and C
 - Value of A influences value of B and C
 - Arrow $\rightarrow$ “direct influence over”
 - A has direct influence over B and C



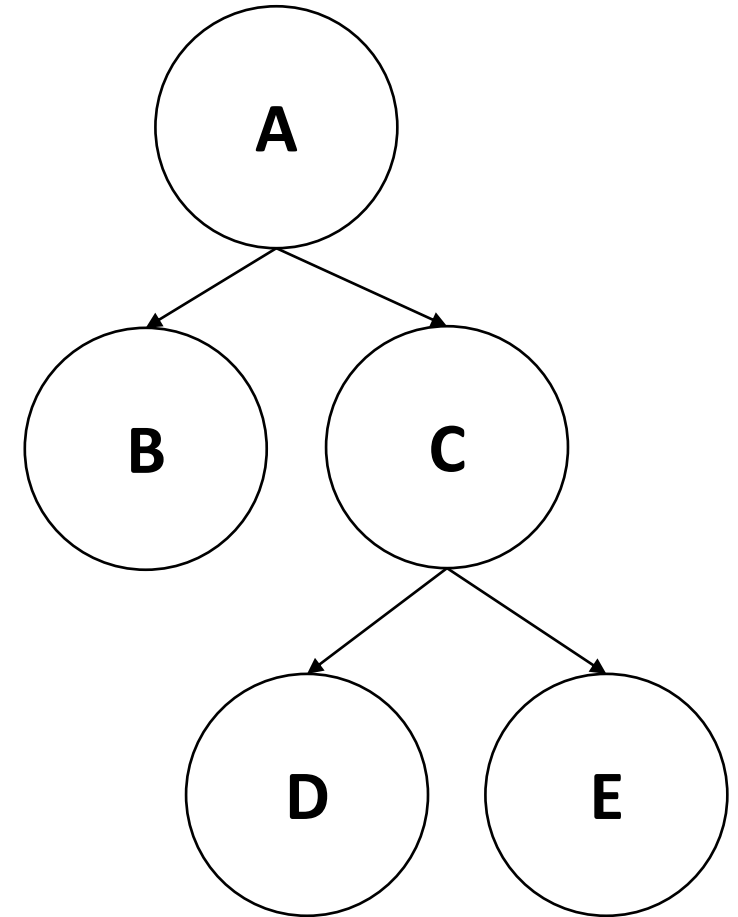
Bayesian Network

- B and C can be dependent
- However, they are conditionally independent given A
- Q: Write down the formula that relates A, B and C

$$P(B, C|A) = ??$$

$$P(B, C|A) = P(B|C, A)P(C|A)$$

$$P(B, C|A) = P(B|A)P(C|A)$$



Bayesian Network

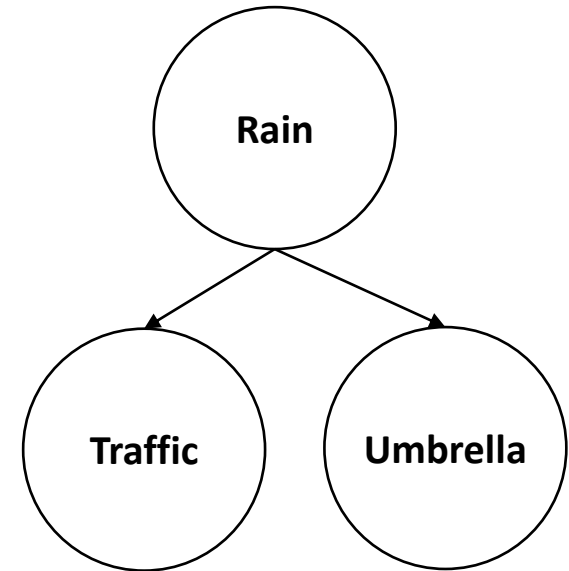
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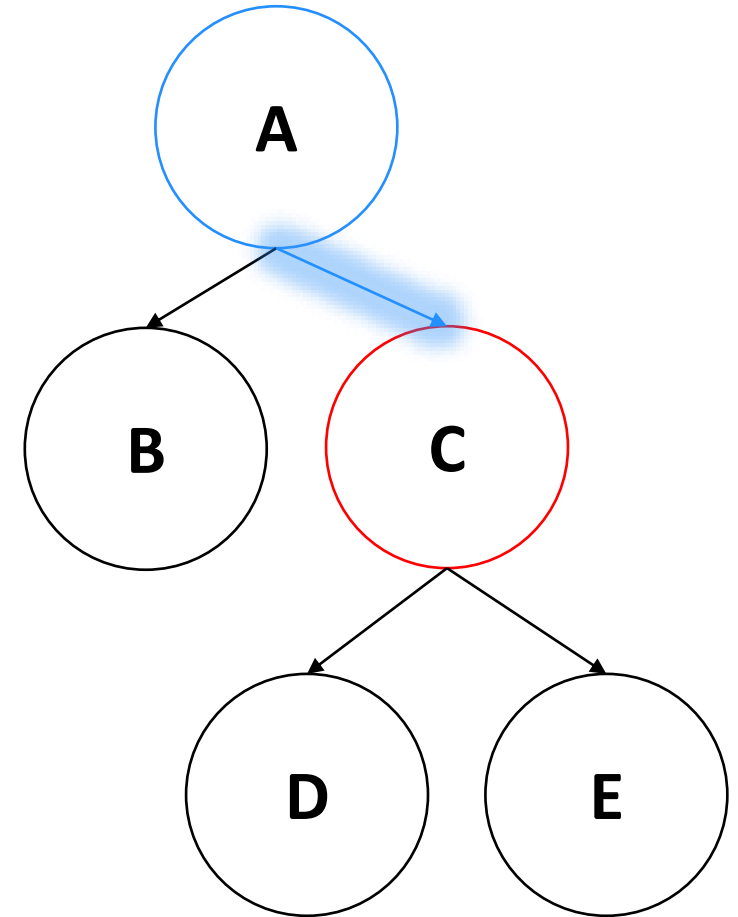
$$P(B, C|A) = P(B|A)P(C|A)$$

- If we do not know A, knowing something about B will tell us something about C (tells us about A which tells us about C)
- but if we know A then knowing B does not tell us anything more about C than we already knew because of A.



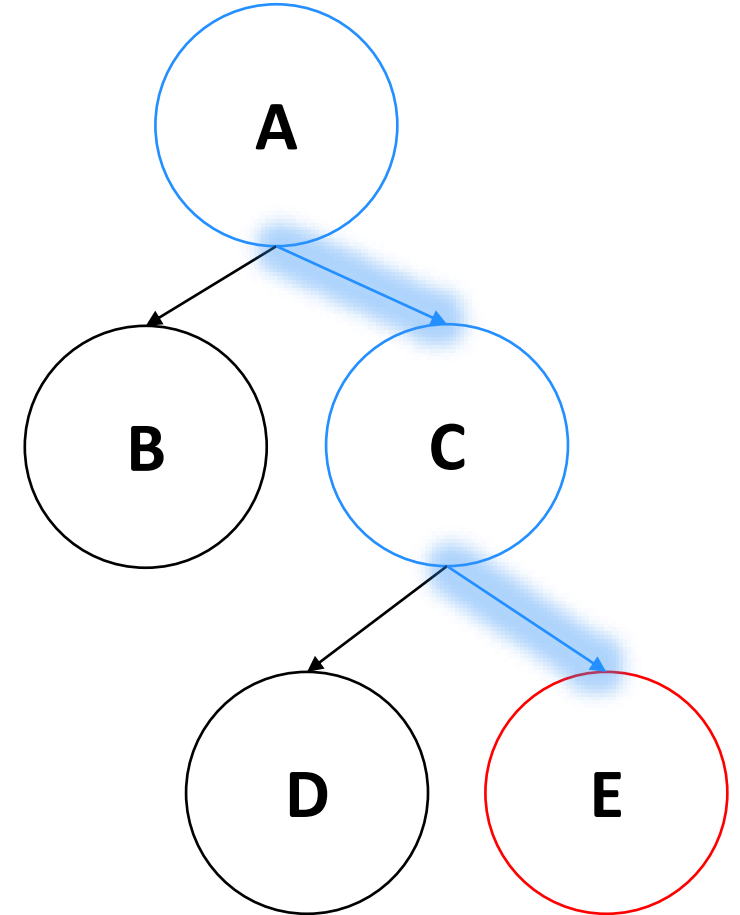
Bayesian network

- C can be dependent on A



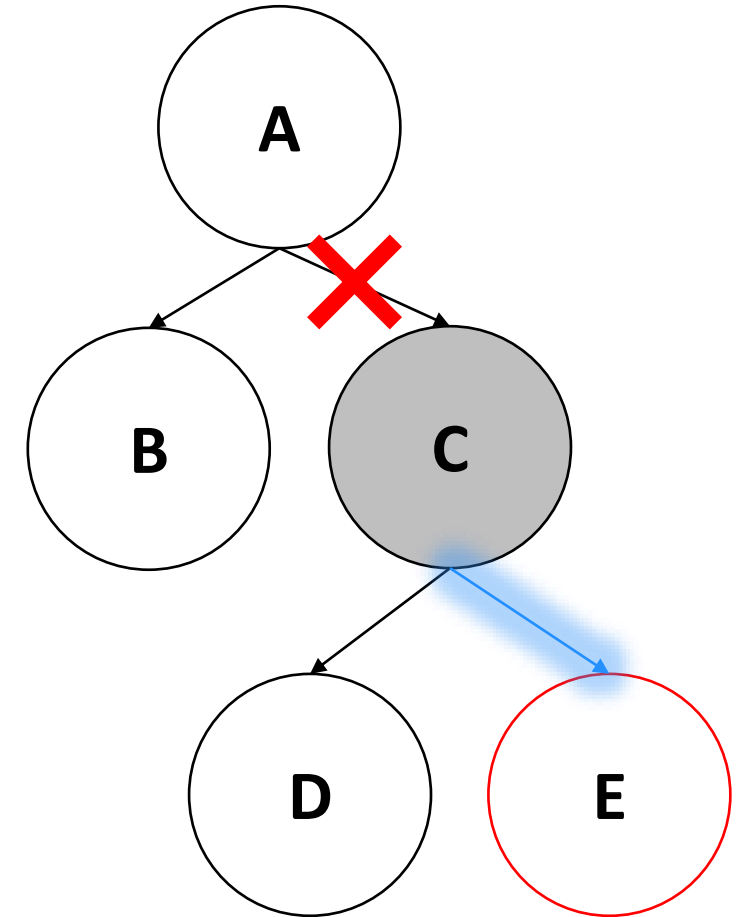
Bayesian network

- C can be dependent on A
- E can be dependent on A and C



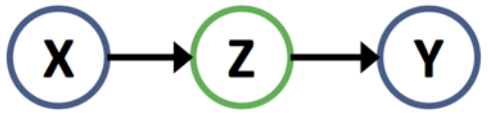
Bayesian network

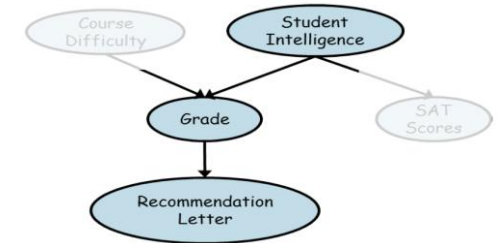
- C can be dependent on A
- E can be dependent on A and C
- However, E is conditionally independent of A given C
 - That is, C captures all the information in A relevant to determine E



Flow of Probabilistic Influence, chain

(Can learning X change our belief about Y?)

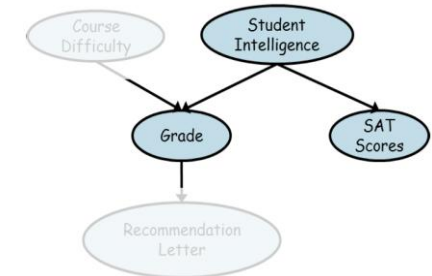
Graphical Structure	NO evidence about Z	YES evidence about Z
	 Can change our belief	 Adds no information



Flow of Probabilistic Influence, common cause

(Can learning X change our belief about Y?)

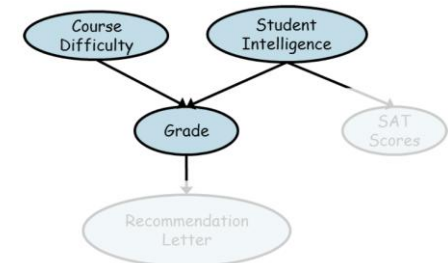
Graphical Structure	NO evidence about Z	YES evidence about Z
		
	 Can change our belief	 Adds no information



Flow of Probabilistic Influence, common effect

(Can learning X change our belief about Y?)

Graphical Structure	NO evidence about Z	YES evidence about Z
		
		
	neither Z nor its descendants observed  Adds no information	or any descendant of Z  Can change our belief



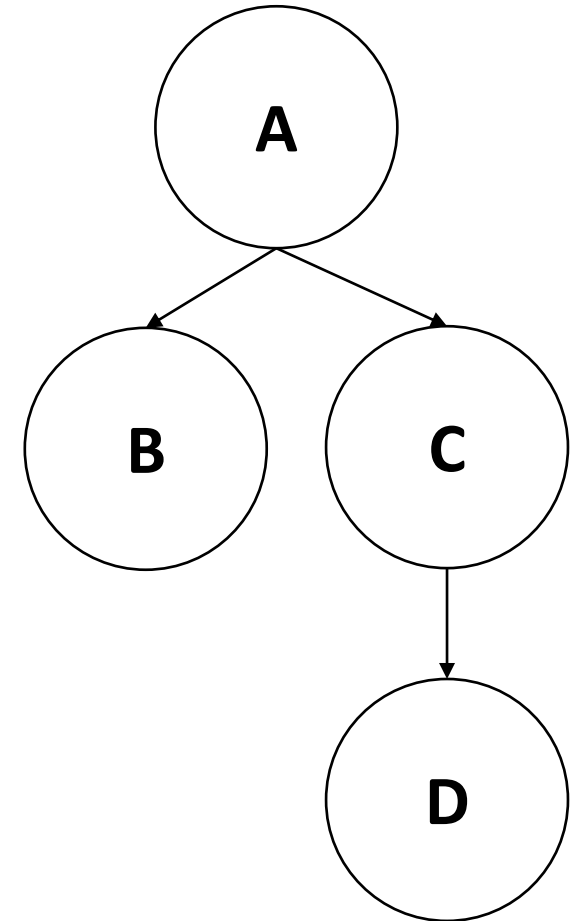
In larger graphs, learning X can change our belief about Y only if at least one path between them is open, given the observed evidence.

Joint Distribution

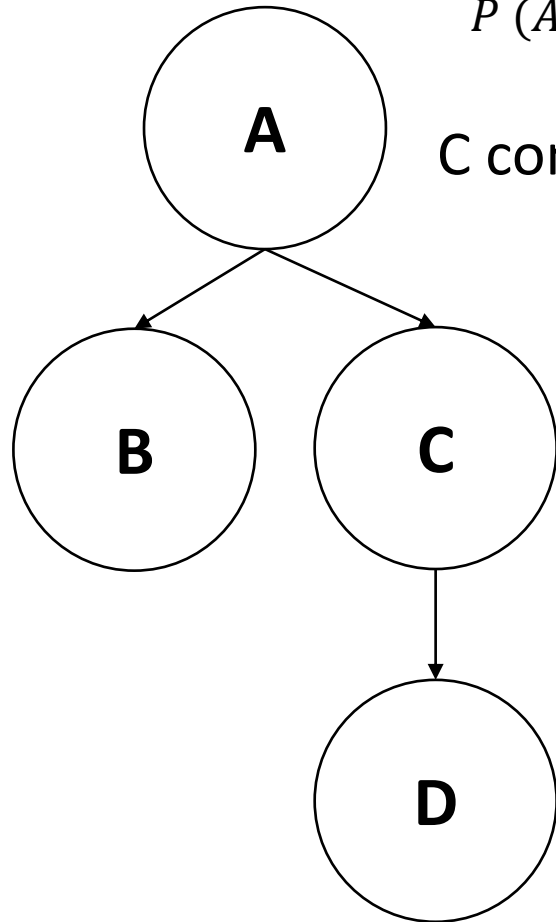
- A probabilistic model of a domain must represent the Joint Probability Distribution (JPD)
 - i.e., the probability of every possible event as defined by the combination of the values of all the variables.
- Bayesian networks achieve compactness by factoring the JPD into local, conditional distributions for each variable given its parents.
- Let's factorize $P(A, B, C, D)$
 - Remember the chain rule

$$\prod_{i=1}^n P(X_i | X_1, \dots, X_{i-1})$$

Tip: We should work from the top and factor out A, then B, C and D



Joint distribution example



$$P(A, B, C, D) = P(A) P(B|A) \underbrace{P(C|A, B)}_{\text{C cond. indep. of B given A}} \underbrace{P(D|A, B, C)}_{\text{D cond. indep. of A, B given C}}$$
$$= P(A) P(B|A) P(C|A) P(D|C)$$

In general:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{i=n} P(X_i | \text{Parents}(X_i))$$

- We could have used the chain or product rule in any order
- This factorization directly exploits the conditional independences encoded by the graph

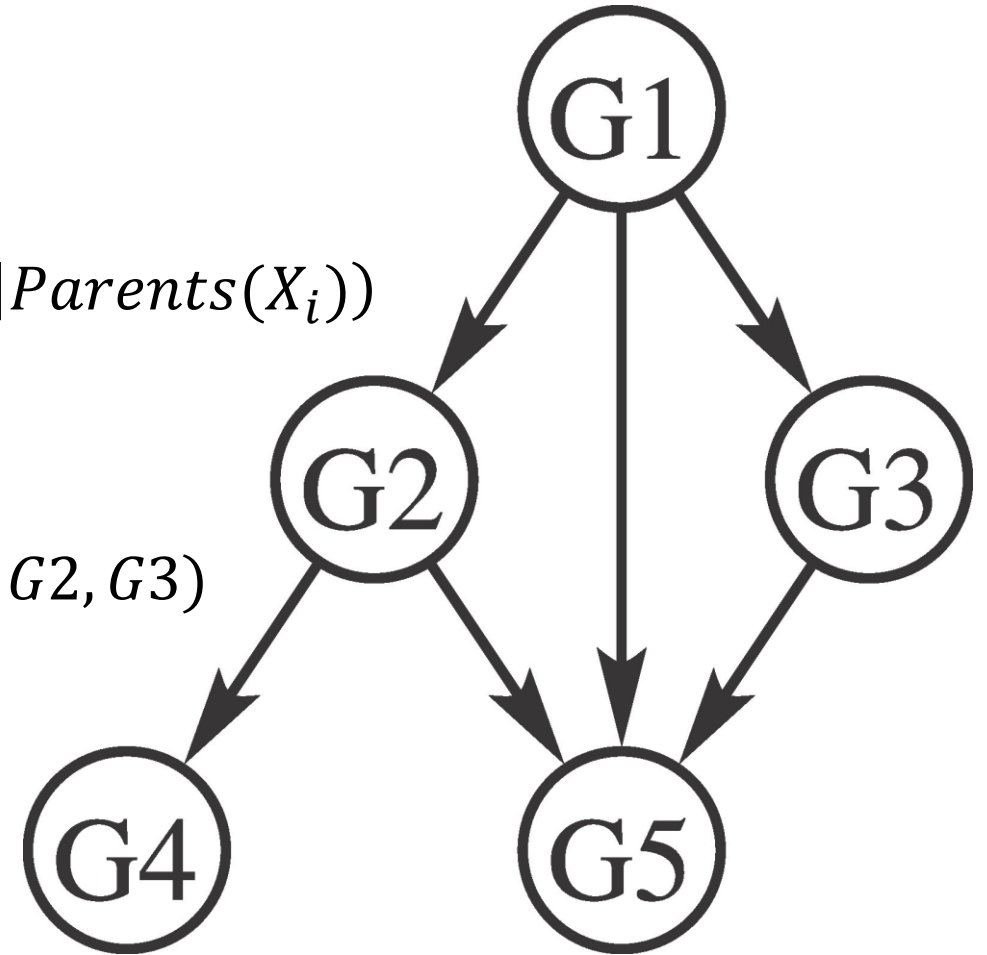
Exercise 2: Factorize the graph

- $p(G1, G2, G3, G4, G5) = ?$

remember

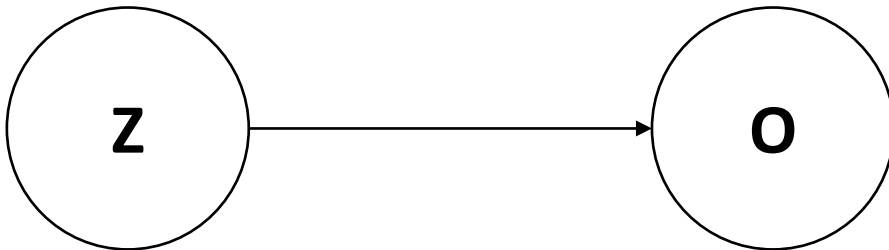
$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{i=n} P(X_i | \text{Parents}(X_i))$$

$$= p(G1) p(G2|G1) p(G3|G1) p(G4|G2) p(G5|G1, G2, G3)$$



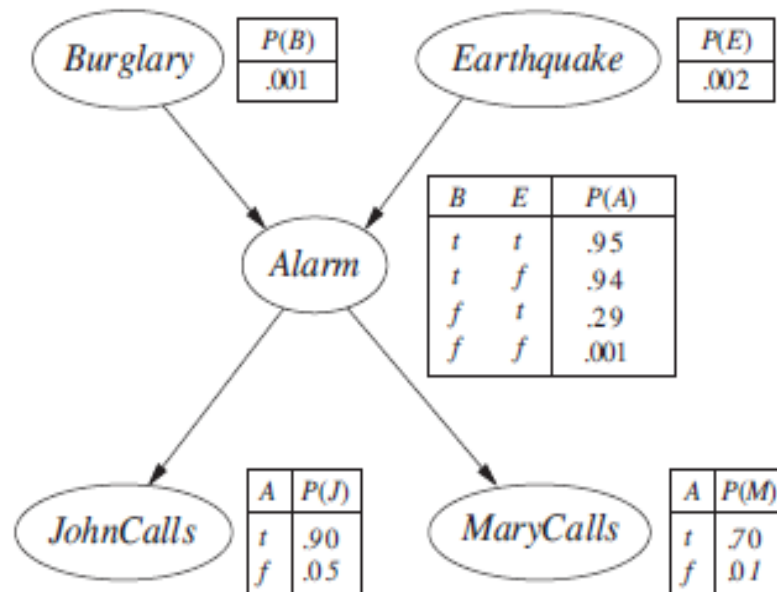
Zebra Example

- Vision system for detecting zebras Z
 - Prior: $p(Z)=0.02$ (zebra in 2% of images)
 - Detector for “stripey areas” gives observations, O
 - Detector gives yes/no
 - Detector performance
 - $p(O|Z) = 0.8$ (true pos)
 - $p(O|\neg Z) = 0.1$ (false pos, e.g. gate)
- Draw as Bayesian network
 - Order? : The existence of the zebra influences the observation and not vice versa



Alarm example

- You have an alarm.
 - Reacts to burglaries but sometimes triggered by small earthquakes.
 - 2 neighbors John, Mary call when they hear the alarm (not earthquake or burglary!)
 - John calls almost every time there is an alarm but can confuse it with a phone
 - Mary plays loud music and sometimes misses it but rarely mix other things with it
 - Draw Bayesian network! What variables?



- Simplification
 - J&M calling does not depend on what triggered the alarm
- Conditional Probability Table
 - Values not specified in the text before. Made up here.

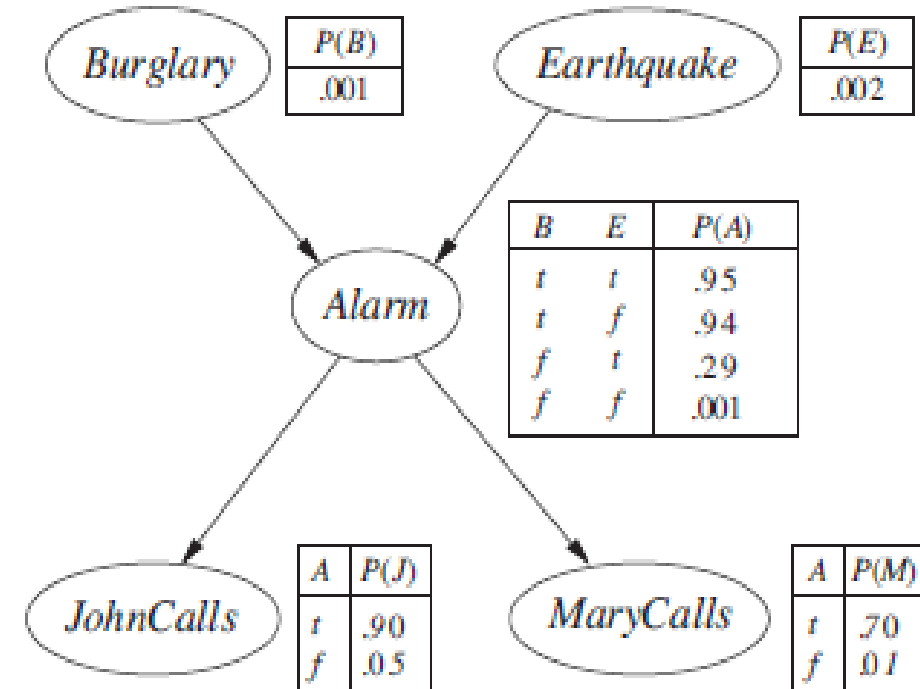
Final Exercise: Alarm example calculation

- Calculate $P(J, M, A, \neg B, \neg E)$

- J : JohnCalls
- M : MaryCalls
- A : Alarm
- B : Burglary
- E : Earthquake

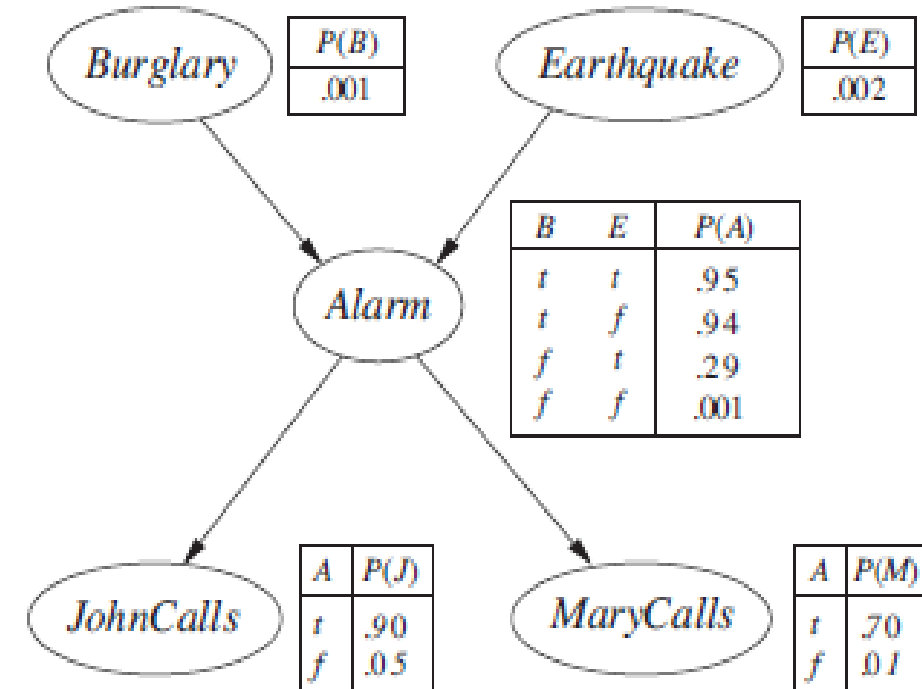
- Remember that $p(\neg X) = 1 - p(X)$ and

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^{i=n} P(X_i | \text{Parents}(X_i))$$



Final Exercise: Alarm example calculation

- Calculate $P(J, M, A, \neg B, \neg E)$
- J : JohnCalls
- M : MaryCalls
- A : Alarm
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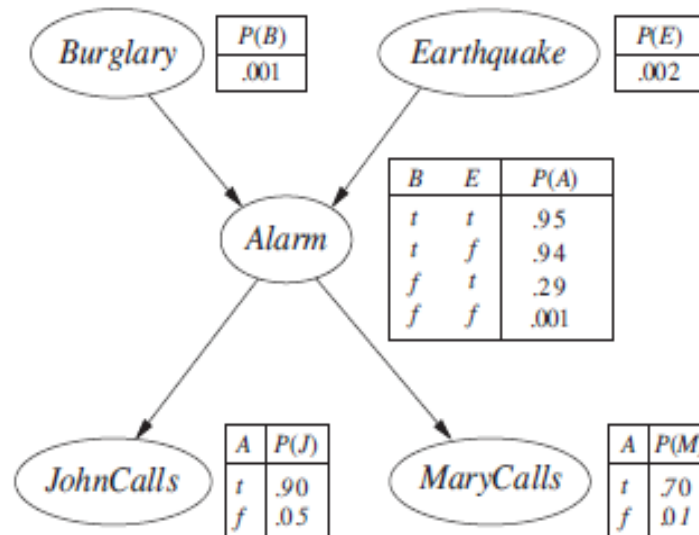


$$P(J, M, A, \neg B, \neg E) = P(\neg B)P(\neg E)P(A|\neg B, \neg E)P(J|A)P(M|A) = 0.0006$$

0.999 0.998 0.001 0.90 0.70

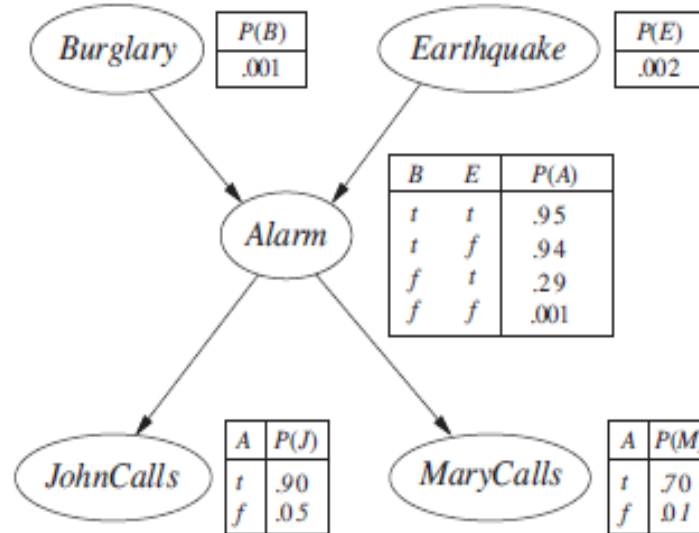
Learning the Model

- In the Alarm network, how do we know that $P(\text{Alarm}|\text{Burglary}, \neg\text{Earthquake})=0.94$?
 - We learn it from data!
 - This is the core of probabilistic machine learning:
 - Parameter Learning: If we know the graph structure (the arrows), we can estimate the Conditional Probability Tables (CPTs) from observed data. For example, we could look at thousands of records and see how many times the alarm went off when there was a burglary but no earthquake.
 - Structure Learning: If we don't know the structure, we can use algorithms to search for the graph that best explains the data.



Tip

- Choosing a Bayesian network structure
 - When causal relationships are known, a cause $\rightarrow$ effect ordering is often a useful starting point.
 - This often gives a compact network with conditional probabilities that are easier to specify.
 - Example: $P(\text{MaryCalls} \mid \text{Alarm})$ describes how Mary responds to an alarm. $P(\text{Alarm} \mid \text{MaryCalls})$ also depends on how frequently alarms occur.



The direction of the arrows does not restrict inference: we can still infer causes from observed effects.

Key points on Bayesian Networks

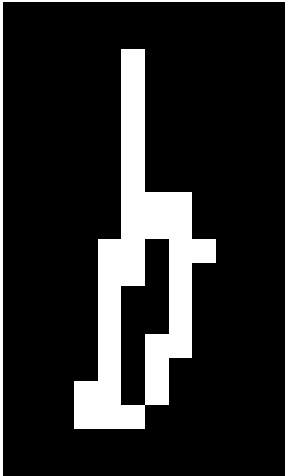
- Representation of the joint probability distribution
 - Through factorization, Bayesian Networks provide a compact representation of the joint probability distribution over a set of variables. This makes understanding and constructing the joint distribution more manageable.
- Encoding of a collection of independence statements
 - The graph encodes conditional independence assumptions. Each variable is conditionally independent of its non-descendants given its parents. More generally, the graph structure and its directed paths determine which independences are guaranteed to hold across the network.
- Helpful in designing inference procedure
 - Knowing the structure and the conditional probability distributions, one can perform various inferential tasks using a Bayesian Network, such as querying the probability of a particular variable given evidence.
 - We will see this in the next lecture with HMMs

Reasoning over Time or Space

- Often, we want to **reason about a sequence** of observations
 - Speech recognition
 - Robot localization
 - User attention
 - Medical monitoring
 - ...
- Need to introduce time (or space) into our models

Sequential data – Example 1

- Measurement of time series
- Example: Sign recognition
 - Measure: drawn path
 - Want: characters



Sequential data – Example 2

- Measurement of time series
- Example: Speech recognition
 - Measure: audio signal
 - Want: Words/sentences



Next Lecture



End of Taming Uncertainty Part 1/2

